

Towards an Automated Coverage Evaluation of the Invasive Quagga Mussel using Benthic Imaging Systems

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Status Quo

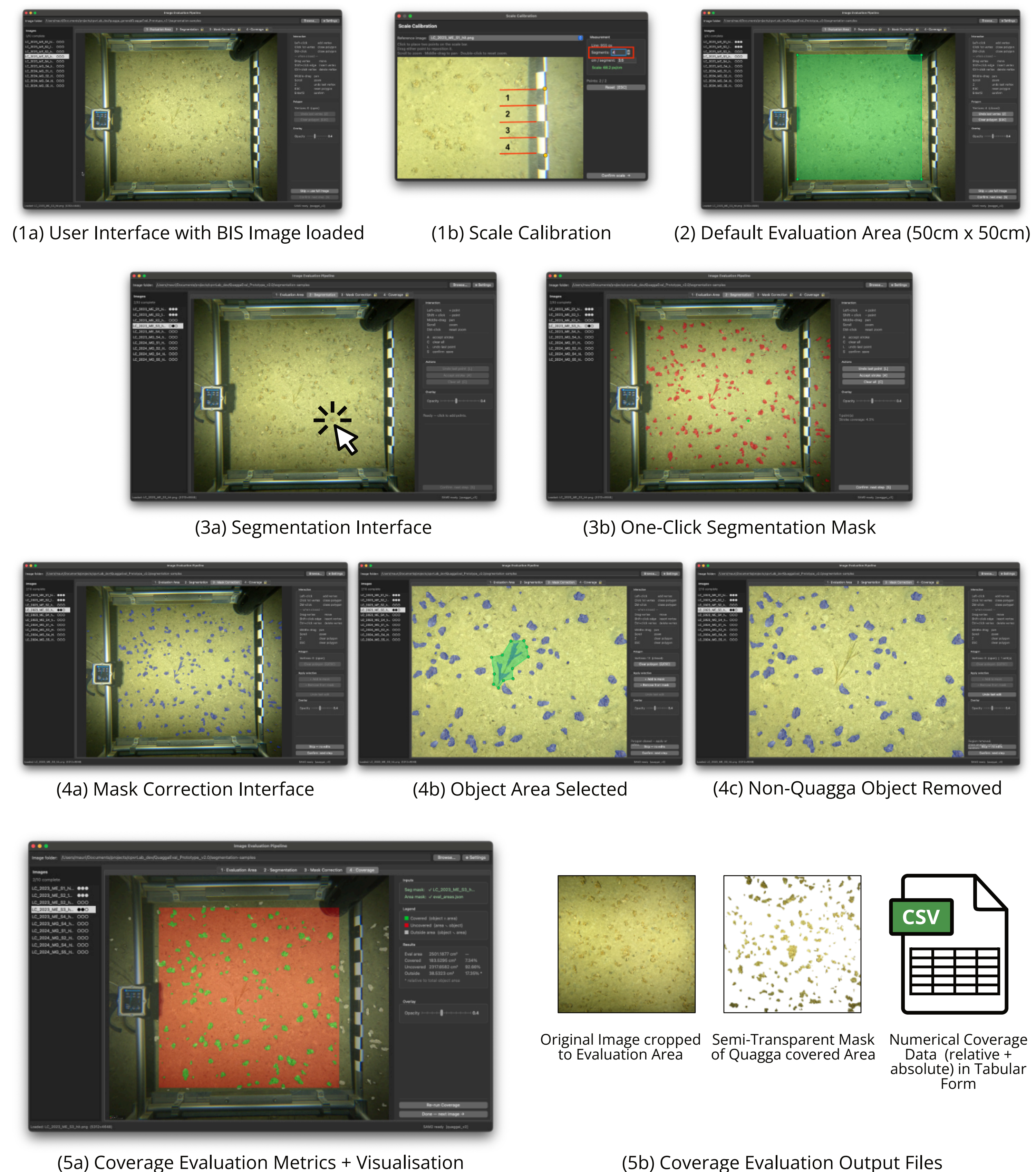
Quagga mussels destroy limnic ecosystems and cause damage to adjacent infrastructure [1, 2]. Systematic monitoring allows to evaluate the current state of the infestation as well as the effectiveness of counter-strategies. "Benthic Imaging Systems" (BIS) are used to capture lake-bed images to monitor the spread of invasive mussels by area covered. The current standard operating procedures (SOP) require manual image evaluation with tools like Adobe Photoshop. This is costly and requires significant amounts of time per evaluated image [3]. Additionally, this approach is prone to subjectivity.

Research Question

"How can we analyse BIS image data from quagga mussel monitoring more efficiently while ensuring consistent and comparable output?"

Software Implementation

For the objective of developing a more efficient monitoring solution, an iterative prototyping approach was used: Insights from ongoing exchanges with eawag [4, 6] and the feedback from users testing intermediary prototypes continuously informed development.



The software works in the following way: Initially the scale is calibrated for a given folder of BIS images (1). Based on the calibration a default evaluation area is selected on each image (2). The user clicks on any quagga mussel in the image and our segmentation model automatically generates a so called "segmentation mask" for all pixels containing quagga mussels (3). This mask can then be corrected manually by selecting areas to be added or removed (4). Coverage evaluation metrics are then calculated, stored in output files and shown to the user (5). Afterwards, the process repeats for the next image at step (2).

About Us

Maurizio Piu, BSc Computer Science

Maurizio is a Master Student in Engineering at the Bern University of Applied Sciences. He works on sensible applications of deep learning methods, particularly in the field of computer vision.



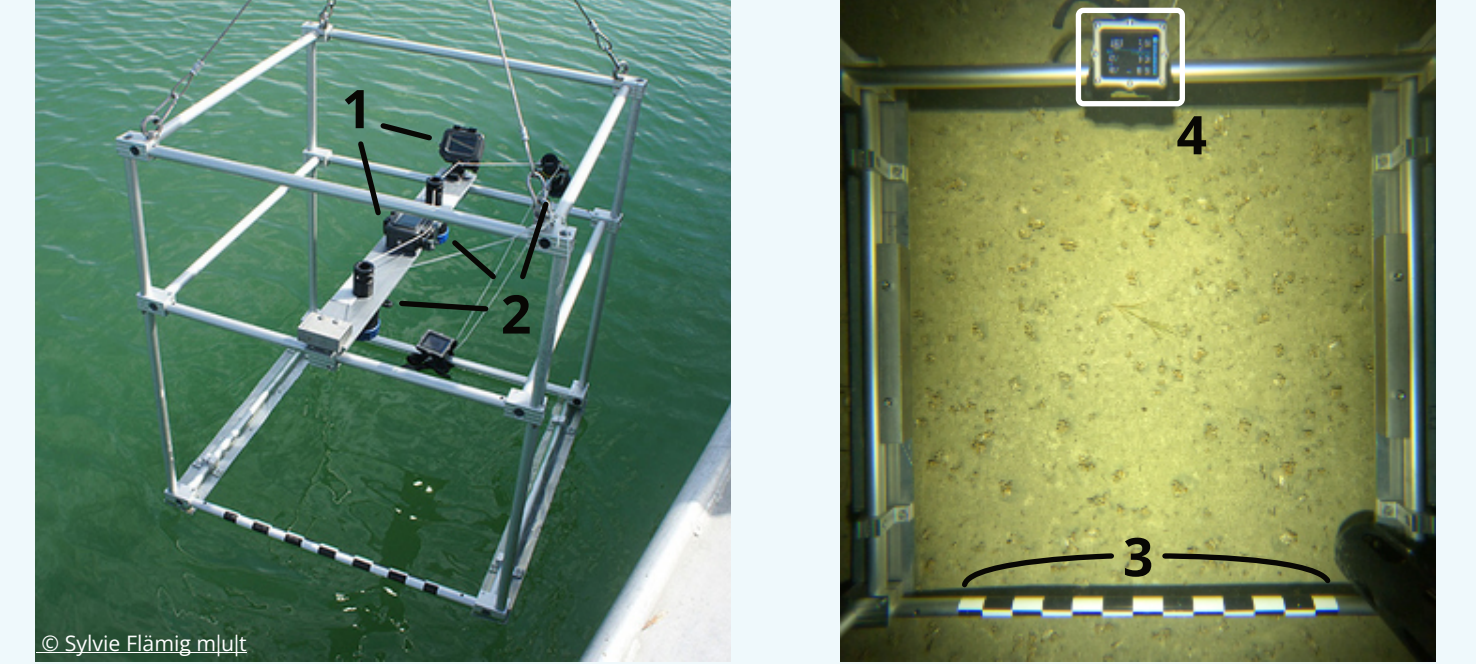
Prof. Dr. Christoph Paulus

Christoph is a professor of computer science at Bern University of Applied Sciences and heads the Computer Vision Laboratory – which also focuses on simulations and artificial intelligence.



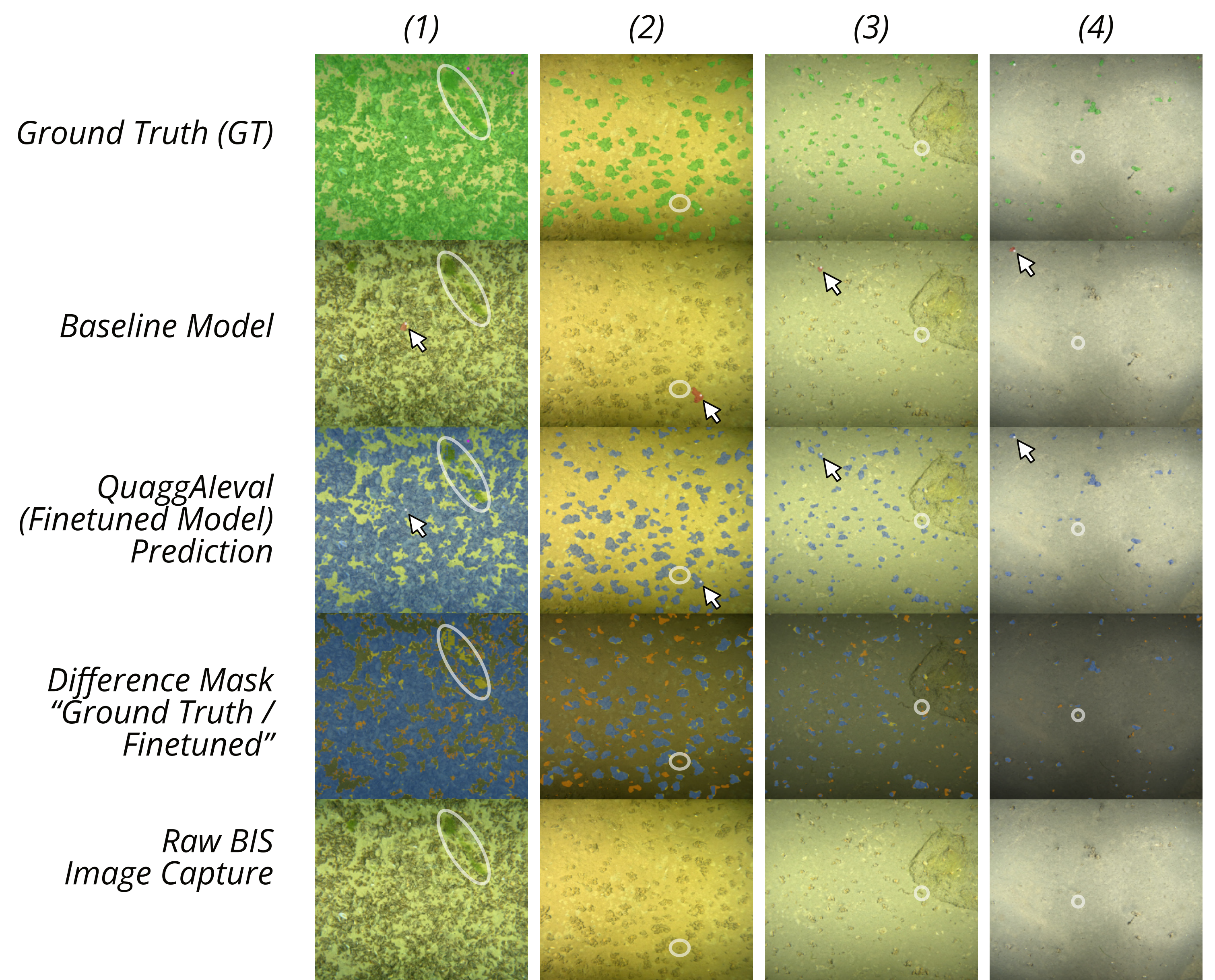
Benthic Imaging System (BIS)

The benthic imaging system (BIS) is used for image-based quagga mussel monitoring. It consists of underwater cameras (1) and diving lamps (2) affixed to a metal frame. Additionally, there is a size reference (3) and dive computer (4) attached to the bottom of the frame. [3]



Finetuning

The core improvement to the monitoring workflow is based on a custom, fine-tuned version of FAIR's SAM2 model [4]. This image segmentation model served as a baseline used for conducting additional training to recognise quagga mussels on BIS image data specifically. The training dataset consists mainly of monitoring data from Lake Constance and Lake Geneva from 2021-2025 and was provided by eawag.

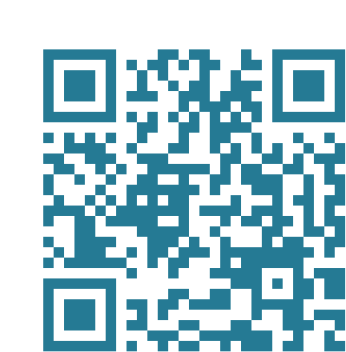


The samples above show the quality of the "single-click" detection. In the difference mask, false positive (not GT but predicted) and false negative errors (GT and not predicted) are visualised in red and yellow respectively. Sample (1) highlights an incorrect "false negative": the predicted mask is *more* precise than the GT mask. Similarly in sample (2) some quagga mussels have not been captured in the GT mask but detected by the prediction. Nonetheless there are also cases in which the model falsely identifies areas as quagga mussels (3) or misses areas that should have been identified as such (4). As such errors are unavoidable, there is a manual correction step built into the processing loop.

"QuaggAlev improves the efficiency of quagga mussel monitoring by at least 39 and up to 959 times"



Eawag Quagga Mussel Competence Center



QuaggAlev Software

Results

When using the current SOP for evaluation, the time required to process one image ranges between 2-16 hours of work. With the same person using the prototype for evaluation, this time can be cut down to at most 3 minutes per image, already resulting in a performance increase of 39 to 959 times faster monitoring.

Acknowledgements

Funding:



References

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- [4] N. Ravi et al., "SAM 2: Segment Anything in Images and Videos," arXiv preprint arXiv:2408.00714, 2024. [Online]. Available: <https://arxiv.org/abs/2408.00714>